CHAPTER

15

Specific aspects of cytotoxic and hormonal drug therapies

OUTLINE

Principles of Tumors
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In the middle of the 19th century—at the same time that tumors began to be identified as cellular masses (see Section 2.1)—rapid advances were being made in chemistry. Numerous new compounds were synthesized and tested for biologic and potentially therapeutic effects. In the 1870s, arsenic trioxide was used as an anticancer agent [1]. This was followed by the work of Paul Ehrlich (1854–1915) on antibacterial arsenical compounds, which led in 1909 to the discovery of the first successful antisyphilis drug "Salvarsan" [2].

In 1911, selenium compounds were synthesized and studied as anticancer agents [3,4], with some reports of benefit to patients with inoperable cancers [5]. Their toxicity limited their use as anticancer drugs [6,7].

Little advance was made until the 1950s, when relatively low-toxic analogues of "mustard gas" (see in Section 15.1.1b) were discovered. Soon after, chemical analogues of folic acid, such as methotrexate, and biological agents, especially the *Vinca* alkaloids, such as 6-mercaptopurine, were found to be effective. Occasionally, a chemical was recognized through studies unrelated to cancer chemotherapy research—as was *cis*-platin [8].

By the 1990s, fewer and fewer new biological agents were being found. With new techniques for analyzing structures of target sites on relevant molecules in cells, drugs began to be synthesized to be structurally complementary to those sites, and hence to potentially inhibit those molecules. Generally, the targets have been the enzymatic sites on enzymes of the signaling pathways or receptors on signaling proteins (see in Chapter 4). In addition, new hormonal therapies based on "designed" drugs were developed.

This chapter describes specific aspects of antitumor cytotoxic and hormonal therapies with attention to translational issues in their effects and side effects.

15.1 General

15.1.1 Differences in chemical structures and mechanisms of effects of cytotoxic drugs

Cytotoxic anticancer drugs, like many carcinogens, react with many macromolecules in the cell and, as discussed in Appendix 8, are subject to tissue- and cellular defensive factors before they reach their target(s).

Anticancer drugs range from small molecules such as arsenic trioxide and *cis*-platin through chain structures such as "mustard" alkylating agents to complex multiring structures, such as taxanes. Most of these drugs appear to have "nongenopathic" and "genopathic" effects (Fig. 15.1; Appendices 3 and 7). None of these kinds of effects are predictable on the basis of their chemical natures or structures [9].

It seems difficult to understand how such chemical diversity in drugs can be consistent with cytotoxic reactions involving only DNA. Rather, the diversity may indicate a large number of different "targets" of the anticancer agents in the cell, which might better correspond to the large number of "sites of effect" on the various proteins which process DNA (see polypharmacology Section 15.1.4 below).

This subsection reviews the chemical and therapeutic effects of each of the major kinds of anticancer agent. All images are from PubChem unless otherwise indicated.

(a) Arsenic trioxide

This compound was used as an anticancer agent in the 19th century (see above) and is now used in the treatment of promyelocytic leukemia. It acts on multiple enzymes, especially those metabolizing glutathione in cells [10,11] (Fig. 15.2).

(b) Nitrogen mustards

These are mainly the nitrogen-mustard alkylating agents, which have been studied in the considerable detail since the 1940s [12]. The protoalkylating agent was the WWI poison gas *bis*(2-choroethyl) sulfide, which smells like mustard but has no other relationship to that plant. The gas not only has acute irritant effects on exposed tissues but also causes reversible bone marrow-suppression in exposed individuals. Analogues in which a nitrogen atom replaces the sulfur atom are less acutely toxic but retain cell-killing effects even in small systemic doses. The compounds are called "alkylating agents" because their chemical reactions involve covalent addition of alkyl groups (alkyl adduct formation) to susceptible macromolecules. Extensive work dating from WWII [13] showed that alkylating agents react with proteins. The components of proteins which appeared to be most reactive at physiological pH and low concentrations of agents were cysteinyl derivatives of amino acids, as well as with alpha-carboxyl, aspartyl, glutamyl, and imidazole (histidine) groups [14]. At the time, the corresponding enzymatic and cytoplasmic structural damage was thought to account for the acute toxic effects of these agents, but not for any late effects such as carcinogenesis.

In the 1960s, the mechanisms of cell killing by alkylating agents were often considered to involve direct covalent reactions with DNA, especially the *N*7 site on guanine, as well as lesser reactivities with other sites on guanine and on other bases.

While this idea is repeated in many books as the probable mechanism of action of alkylating agents, there is evidence that the important actions of many active anticancer drugs are not with DNA. For example, chlorambucil (4-[bis(2-chloretyl) amino] benzenebutanoic acid) alkylates DNA but is effective in chronic lymphocytic leukemia, in which cell proliferation is not a marked feature. Chlorambucil may induce apoptosis via actions on non-DNA structures such as proteins in the cell [15] (Fig. 15.3).

(c) Antimetabolites

(i) Methotrexate

This drug is classified as an "antimetabolite" because it interferes specifically with a vital metabolic process in the cell. Methotrexate is an analogue of folate and is taken up into the cytoplasm of cells via the low pH folate transporter or reduced folate carriers, where it then inhibits the specific enzyme dihydrofolate reductase

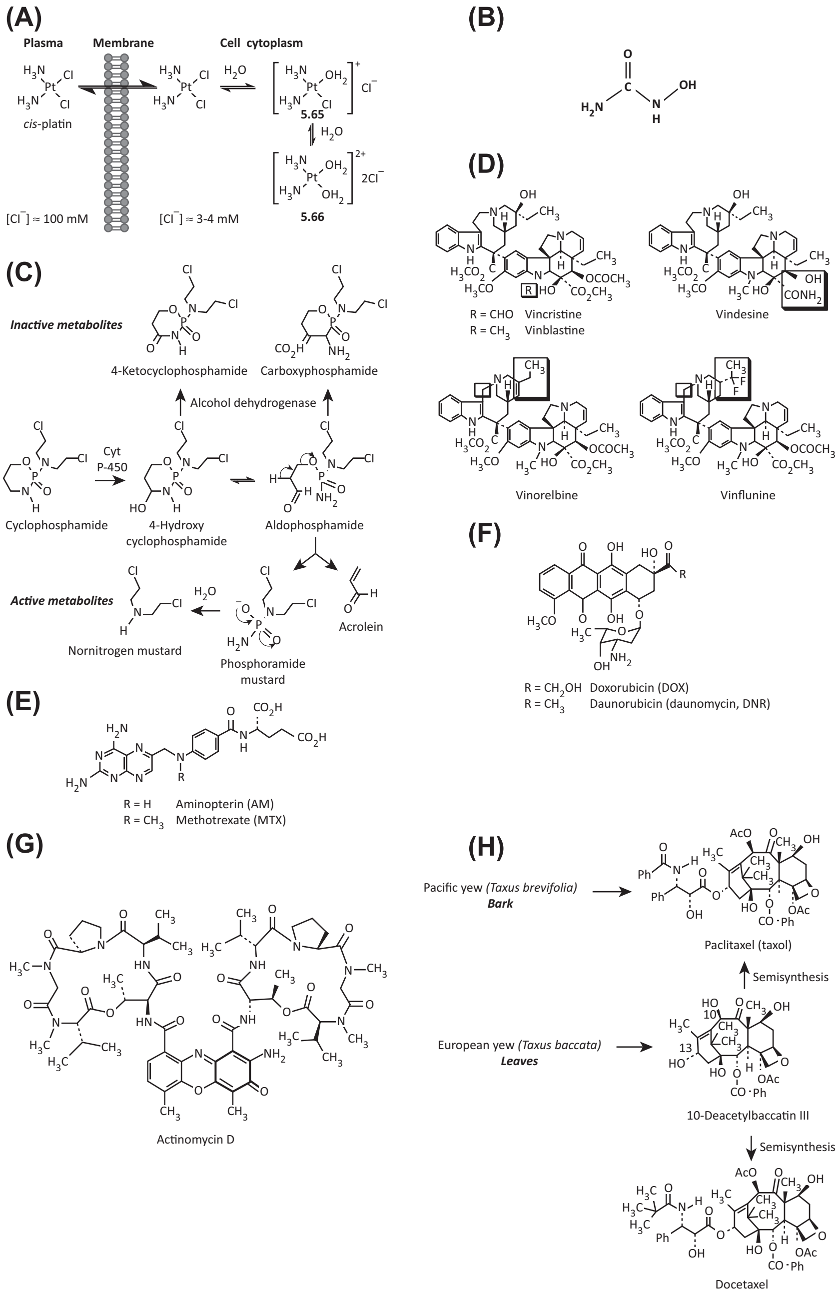

FIGURE 15.1 Structures and activations of some commonly used anticancer drugs. (A) Cis-platin bioactivation. (B) Hydroxyurea. (C) Cyclophosphamide showing bioactivations and inactivations. (D) Vinca alkaloids. (E) Aminopterin and methotrexate. (F) Doxorubicin and daunomycin. (G) Actinomycin. (H) Semisynthesis of taxanes. Some synthetic modifications in the Taxol group of drugs are included. *Source: Chapter 7 — Other anticancer drugs targeting DNA and DNA-associated enzymes. In: Avendaño C, Menéndez JC, editors. Medicinal chemistry of cancer drugs. 2nd ed. Philadelphia: Elsevier; 2015. p. 273–323.*

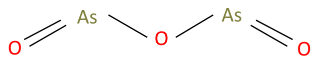

FIGURE 15.2 Arsenic trioxide. PubChem Compound Database; CID = 14888. *Source:* *https://pubchem.ncbi.nlm.nih.gov/compound/14888*.

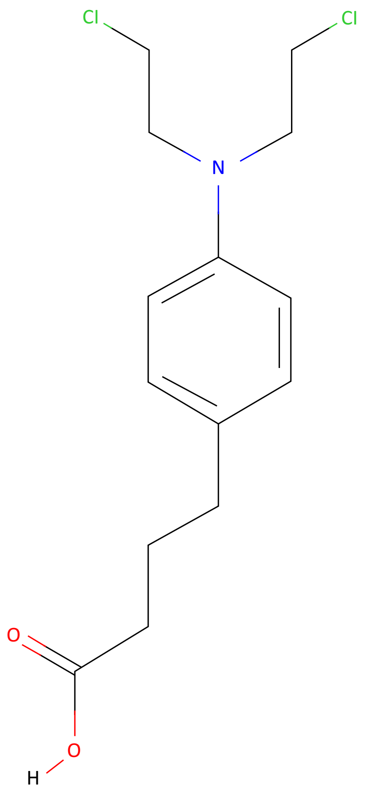

FIGURE 15.3 Chlorambucil. PubChem Compound Database; CID = 2708. *Source:* *https://pubchem.ncbi.nlm.nih.gov/compound/2708*.

[16]. As a result, the cell has reduced capacity to synthesize new purine and thymidine nucleotides, as well as proteins. However, the drug is also thought to inhibit purine metabolism, as well as inhibiting methyl transferase activity, with secondary effects on immune functions [17]. The main side effects of the drug occur in labile tissues (those which constantly produce specialized cells from local tissue stem cells—hematologic, gastrointestinal, see Appendix A1.3.2) Hepatic and renal toxic effects also are common, presumably reflecting defensive factors (Appendix A3.2 and A3.3). There appears to be no selective uptake of methotrexate into tumor cells, and the greater susceptibility of tumor cells to methotrexate over normal cells may relate to lower concentrations of the enzyme in tumor cells (Appendix A4.2.1). Methotrexate does not enter the nucleus but inhibits S-phase DNA synthesis. It has no other genopathic effects, such as induction of chromosomal aberrations [18] (see Appendices A3.1.2 and A7.2) (Fig. 15.4).

(ii) 5-Fluorouracil

5-FU acts in several ways, but principally as a thymidylate synthase (TS) inhibitor. Interrupting the action of this enzyme blocks synthesis of the pyrimidine thymidine, which is a nucleoside required for DNA replication. TS methylates deoxyuridine monophosphate (dUMP) to form thymidine monophosphate (dTMP). Administration of 5-FU causes a scarcity in dTMP, so rapidly dividing cancerous cells undergo cell death via thymineless death [19]. Calcium folinate provides an exogenous source of reduced folinates and hence stabilizes the 5-FU-TS complex, hence enhancing 5-FU's cytotoxicity [20] (Fig. 15.5).

(iii) Hydroxyurea and nitrosoureas

Hydroxyurea has a hydroxyl group on one of the nitrogen atoms of a urea molecule. In the body, the drug scavenges tyrosyl-free radicals, resulting in inhibition of conversion of ribonucleotides to deoxyribonucleotides, and hence the synthesis of DNA [21]. Any agent that affects the enzymes responsible for DNA or RNA synthesis can be included as a "genopathic agent" by the definition used in this book (A3.1.2).

Hydroxyurea may well have other actions, because it has beneficial effects in patients with sickle cell anemia [22]. These effects are

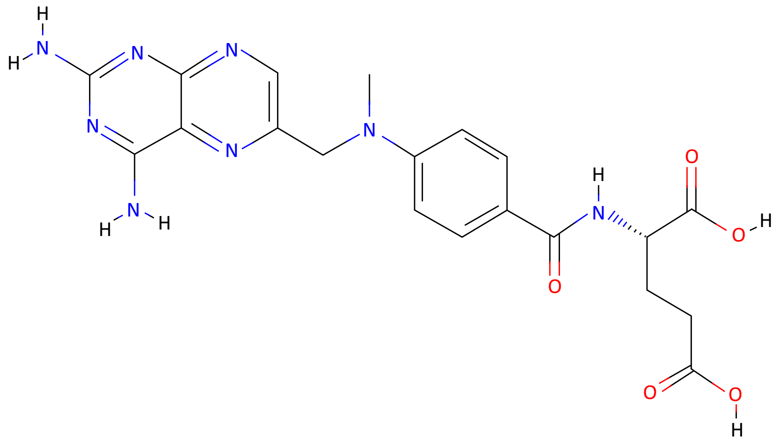

FIGURE 15.4 Methotrexate. PubChem Compound Database; CID = 126941. *Source: https://pubchem.ncbi.nlm.nih.gov/compound/126941*.

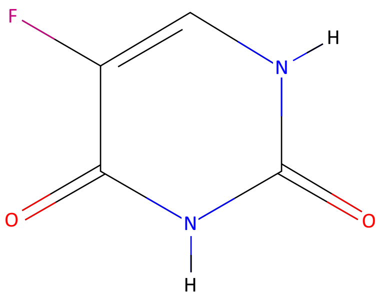

FIGURE 15.5 5-Fluorouracil. PubChem Compound Database; CID = 3385. *Source: https://pubchem.ncbi.nlm.nih.gov/compound/3385*.

unexplained but are unlikely to be caused by inhibiting DNA synthesis.

Nitrosourea drugs have a nitroso group on one nitrogen atom of urea. The various analogues in the group have additional groups on the other nitrogen atom. They react with DNA and are classed as alkylating agents. These drugs pass the blood–brain barrier and are currently one of the "first-line" therapies for glioblastoma [23] (Fig. 15.6).

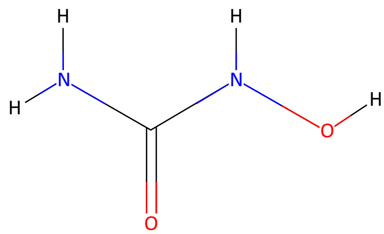

FIGURE 15.6 Hydroxyurea. PubChem Compound Database; CID = 3657. *Source: https://pubchem.ncbi.nlm.nih.gov/compound/3657*.

(d) Cis-platin

Cis-platin is one of the most widely used alkylating drugs used in anticancer therapy. It was discovered in 1965, and clinical trials of its usefulness as an anticancer drug began in the 1970s. It consists of a platinum ion with two methyl and two chloride groups attached. The active drug has the chloride groups adjacent to each other (*cis* form), while the inactive isomer (*trans*-platin) has the chloride ions on opposite sides. Both isomers enter the cell where the chloride ions dissociate and both form adducts on DNA (the position of the methyl groups then being the only difference between the active drug and its inactive analogue). Platin drugs do not enter into tumor cells more than they penetrate normal cells. The greater sensitivity of tumor cells compared with normal cells is not explained [24,25].

Clinically, the drug is superior to nitrogen mustard alkylating agents because, although it causes more nausea and vomiting, it causes less suppression of the bone marrow. However, its side effects are numerous and complex [26].

There are tumor type–specific differences in sensitivity to *cis*-platin, for example, as Zhang et al. [27] point out: "although the platinum-based anticancer drugs *cis*-platin, carboplatin, and oxaliplatin have similar DNA-binding properties, only oxaliplatin is active against colorectal tumors." Extensive comparative toxicokinetic and -dynamic studies in the different tumor types have not been reported.

The mechanism of action of *cis*-platin has been thought to involve cross-linking through the "naked" chloride-dissociation sites on the platinum ion with DNA. To explain how the *trans*-platin can bind to DNA but not kill cells, it has been suggested that *trans*-platin inactivity stems from two major factors:

(i) The kinetic instability promoting its deactivation and
(ii) The formation of DNA adducts characterized by a regioselectivity and a stereochemistry different from those of *cis*-platin [28].

It is known that barely 1% of nuclear *cis*-platin is bound to DNA, the remainder to unspecified nuclear proteins. It is clearly possible that *cis*-platin acts also on proteins associated with processes relevant to the genome, but little information by way of detailed studies of this possibility, especially in comparison with its inactive *trans*-analogue, is available (Fig. 15.7).

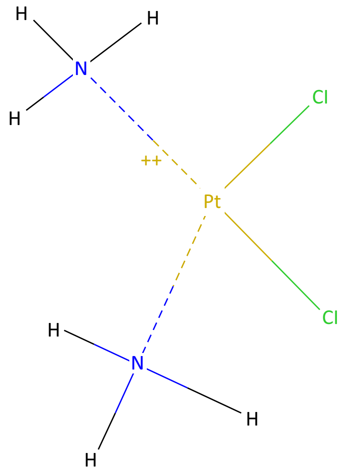

FIGURE 15.7 Cis-platin. PubChem Compound Database; CID = 441203. *Source: https://pubchem.ncbi.nlm.nih.gov/compound/441203.*

(e) Poly (ADP-ribose) polymerase inhibitors

PARP1 is a protein that is important for repairing frequently occurring single-strand breaks ("nicks") in the DNA which occur in the cell cycle. If such nicks persist unrepaired until DNA is replicated (which must precede cell division), then the replication itself can cause double-strand breaks to form [29]. Drugs that inhibit PARP1 cause multiple double-strand breaks to form in this way, and in tumors with BRCA1, BRCA2, or PALB2 [30] mutations, these double-strand breaks cannot be efficiently repaired, leading to the death of the cells. Normal cells that do not replicate their DNA as often as cancer cells and that lack any mutated BRCA1 or BRCA2 still have homologous repair operating, which allows them to survive the inhibition of PARP [31,32].

PARP inhibitors lead to trapping of PARP proteins on DNA in addition to blocking their catalytic action [33]. This interferes with replication, causing cell death specifically in cancer cells, which grow faster than noncancerous cells.

Some cancer cells that lack the tumor suppressor PTEN may be sensitive to PARP inhibitors because of downregulation of Rad51, a critical homologous recombination component, although other data suggest PTEN may not regulate Rad51 (Ref. [33]). Hence, PARP inhibitors may be effective against many PTEN-defective tumors (e.g., some aggressive prostate cancers).

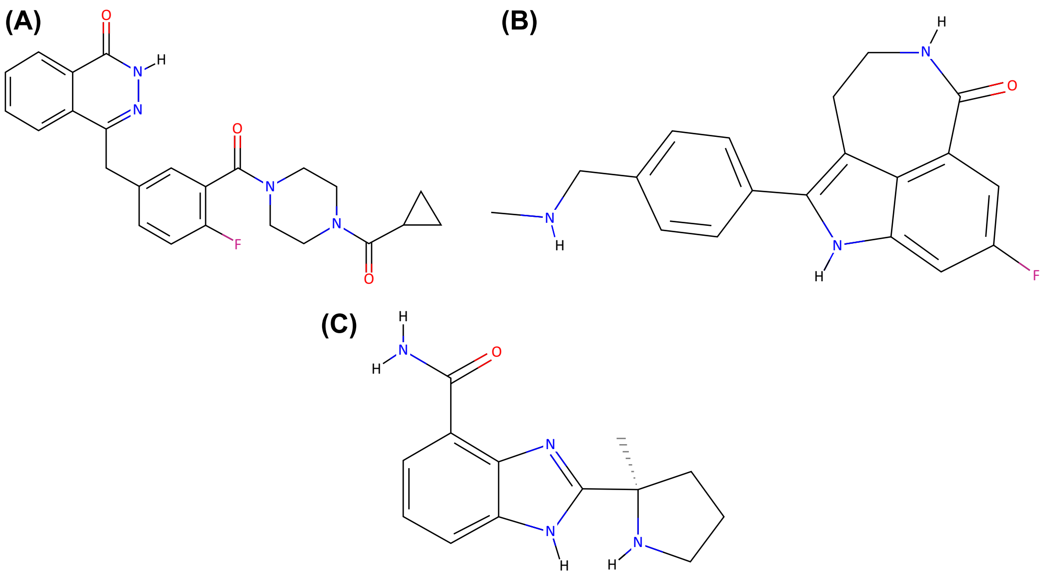

FIGURE 15.8 A PARP inhibitors. (A) Olaparib. PubChem Compound Database. (B) Rucaparib. (C) Veliparib. *Source: (A) https://pubchem.ncbi.nlm.nih.gov/.*

Cancer cells that are low in oxygen (e.g., in fast-growing tumors) are sensitive to PARP inhibitors [34–36] (Fig. 15.8).

(f) Antimicrotubule agents

(i) *Vinca* alkaloids

These drugs do not react with DNA, but rather affect proteins involved in mitosis, especially of the centrosome and the mitotic spindle. These drugs either prevent microtubule formation or prevent the dissociation of microtubules to individual tubulin molecules [37]. Examples of the first of these groups are the *Vinca* alkaloids, vincristine, and vinblastine (Fig. 12.4D). These were discovered in the 1950s and found to have clinically useful antitumor activities. These drugs are now known to react with soluble tubulin in the cytoplasm of cells. Vincristine and vinblastine disrupt the microtubules which form part of the spindle of fibers used by the chromosomes to separate in telophase. Mitoses therefore are disrupted, and cells die by "mitotic death" or nuclear disruptive events. However, these drugs also affect other proteins and have been reported to interfere with DNA, RNA, and lipid synthesis [38] (Fig. 15.9).

(ii) Taxanes

The principal mechanism of action of the taxane class of drugs is the disruption of microtubule function. Microtubules are essential to cell division, and taxanes stabilize GDP-bound tubulin in the microtubule, thereby inhibiting the process of cell division as *depolymerization* is prevented. Thus, in essence, taxanes are mitotic inhibitors. In contrast to the taxanes, the vinca alkaloids prevent mitotic spindle formation through inhibition of tubulin *polymerization*. Both taxanes and vinca alkaloids are, therefore, named spindle poisons or mitosis poisons, but they act in different ways. Taxanes are also thought to be radiosensitizing [39] (Fig. 15.10).

(iii) Other

These are mainly epothilones and discodermolides [40] and they "stabilize" (prevent the

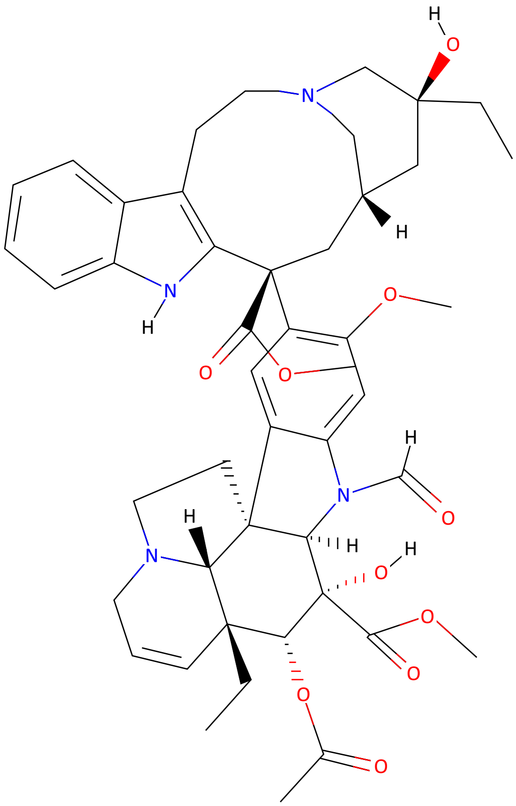

FIGURE 15.9 Vincristine. PubChem Compound Database; CID = 5978. *Source:* *https://pubchem.ncbi.nlm.nih.gov/compound/5978*.

physiological dissociations of) the microtubules and arrest cells in the G2-M part of the cell cycle.

(g) Antitumor drugs related to antibiotics

Actinomycin D (dactinomycin) (Fig. 12.4G) was isolated from *Actinomyces antibioticus* in the 1940s, during searches for antibacterial substances in microorganisms. Actinomycin D was found to have antitumor activities, but no useable antibacterial potency. It is commonly used in anticancer therapies against a limited range of relatively rare tumor types, including choriocarcinoma of the placenta, Wilm's tumor, and Ewing's sarcoma [41]. Its main side effects are against labile cell

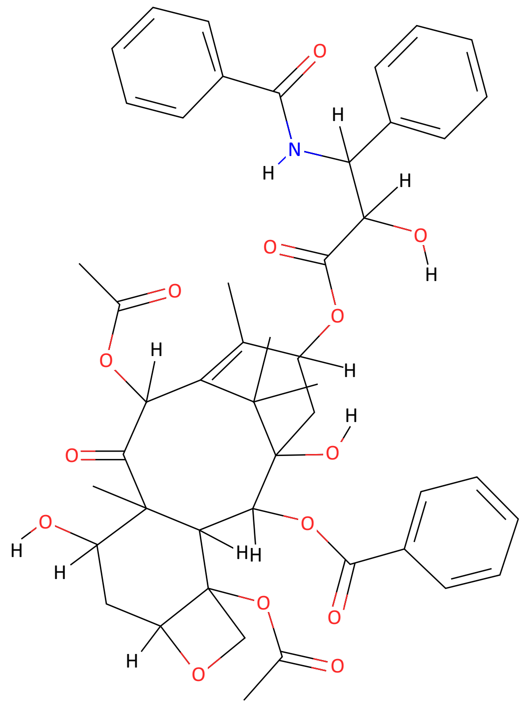

FIGURE 15.10 Taxol. PubChem Compound Database; CID = 441203. *Source:* *https://pubchem.ncbi.nlm.nih.gov/compound/36314*.

types, such as the hematologic, gastrointestinal, cutaneous, and immune systems. The drug causes little chromosomal damage (i.e., a weak clastogen, see Appendix A3.1.4) and is not known to be carcinogenic other than perhaps to cause leukemia through topoisomerase inhibition (see mention of etoposide, below).

The mechanism(s) of action of actinomycin D are not fully understood. Its effect may be via inhibition of DNA-dependent synthesis of RNA (i.e., RNA polymerases; transcription) through its noncovalent binding to specific DNA motifs near transcriptional complexes. It may bind the minor groove of DNA and/or have an "interfacial" (i.e., intercalatory, noncovalent) action on the DNA-RNA polymerase sites [42].

Other antitumor antibiotics may act on genomic process—related proteins. For example, the anthracyclines (doxorubicin and daunorubicin) (Fig. 12.4E) probably mainly act on the functions of topoisomerases in cells by forming a stable DNA–drug–enzyme complex [43]. Etoposide (an analogue of podophyllotoxin) acts on topoisomerase II, especially when the enzyme is complexed with DNA. The drug reacts only weakly with DNA in the absence of the enzyme (see also next subsection). The mechanisms by which etoposide might cause chromosomal aberrations is discussed in Ref. [44] (Fig. 15.11).

15.1.2 Activation of prodrugs to active compounds

Just as for some chemical carcinogens (Section 3.3.1), some drugs are given as prodrugs, which are activated in the body. The activating enzymes are mainly members of the CPY/P450 family, members of which activate carcinogens, and also deactivate toxins (Section 13.2.3; Appendix A8.1.3).

Drugs activated from a proform include antimetabolites such as 5-fluorouracil (5-FU) and mercaptopurine; alkylating agents such as melphalan and cis-platin; antibiotics such as mitomycin C; as well as others such as cyclophosphamide (Fig. 12.4B), etoposide, and paclitaxel (Fig. 12.4F). The enzymes involved are mainly the P450 cytochromes and other oxidoreductases and transferases [45].

15.1.3 Differences in the potencies for a variety of biological effects among different analogues in the same chemical class

Just as for carcinogens (see Section 3.3.3), wide differences in potencies occur between analogues of anticancer drugs (Ref. [12]). The differences in efficacies are often considered to be based on different degrees of effectiveness of defensive factors in the tissues and cells against the analogues (see Appendix 8). However, many biologically inactive analogues are known

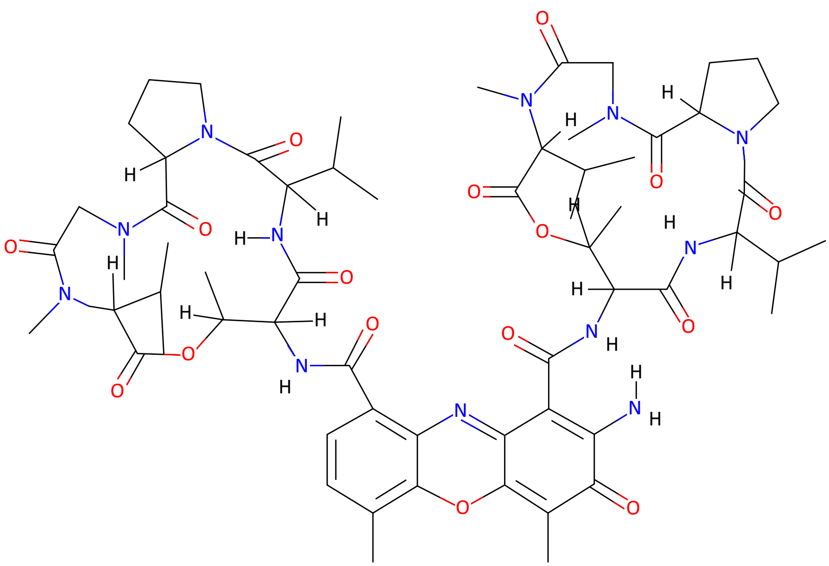

FIGURE 15.11 Actinomycin D. PubChem Compound Database; CID = 441203. *Source: https://pubchem.ncbi.nlm.nih.gov/compound/457193.*

to reach the genome (evidenced by forming adducts on DNA, see discussion of *cis*-platin above). Hence, it would seem that exquisite structural sensitivity in some non-DNA target could be involved, as is suggested in Appendix 8, for carcinogenic analogues versus noncarcinogenic analogues.

Predicting biological activity from chemical structures is recognized to be complex [46,47].

15.1.4 Multiplicity of molecules affected: "polypharmacology"

As in the explanation of the actions of carcinogens (see Appendix A3.3), the fact that many individual drugs have different cell biological effects in different normal cells suggests that these drugs may affect multiple different "targets" and presumably have more than one mode of action [48].

Alkylating agents, for example, are simply highly reactive substances capable of binding to a variety of molecules and thus are liable to have many effects, which probably depend in detail on pharmacokinetic factors. This phenomenon is now fully recognized in the term "polypharmacology." It represents a challenge in drug development, which may be overcome by better drug designs [49,50].

15.1.5 Techniques for increasing diffusion and active transport of drugs into tumor cells

As with all drugs, the pharmacokinetic problem of reaching the cell and subsequently reaching the target on or in a cell applies to anticancer drugs. To reach the cell from the blood stream, drugs must be water soluble or in a water-carried structure such as a micelle to reach the cells. Mechanisms of passage of drugs through the cell membrane are thought to be mainly (i) simple diffusion, (ii) facilitated diffusion, (iii) active transport using receptor binding, and (iv) cell membrane–initiated vesicles (endocytosis, older term "pinocytosis," see next subsection). The relative importance of passive and facilitated diffusion, as well as active transport mechanisms, is unclear [51,52].

15.1.6 Endocytosis-dependent drug uptake into cells

Endocytosis is a mechanism by which cells internalize plasma membrane, surface receptors and their ligands, viruses, and various extracellular soluble molecules. It has been considered the mechanism by which cells clear receptors from their surfaces and become less responsive to corresponding stimuli. More recently, endocytosis has been suggested to be relevant to cellular homeostasis and control of proliferation and accordingly, when disturbed, play a role in hyperproliferative conditions such as cancer [53,54] (Fig. 15.12).

For drug delivery purposes, antisurface receptor antibodies can be "tagged" with a cytotoxic drug, and so enters the cell when the antibody–membrane molecule complex is cleared by endocytosis. Based on the same principle, endocytosis is also thought to be potentially useful for transferring supramolecular structures containing drugs into cells. These supramolecular structures include liposomes [55], nanoparticles, and polymersomes—self-assembling lipid polymers [56].

Nanoparticles are constructed from a wide diversity of chemical materials, and few generalizations about their properties as vehicles are possible [57].

Biological challenges include (i) limited access to dormant cells or micrometastases and (ii) the antiapoptosis characteristic of cancer stem cells [58].

A further issue is whether or not the endocytosed supramolecular structure actually releases the drug into the cell (i.e., preserves bioavailability). There is only limited literature on this point [59].

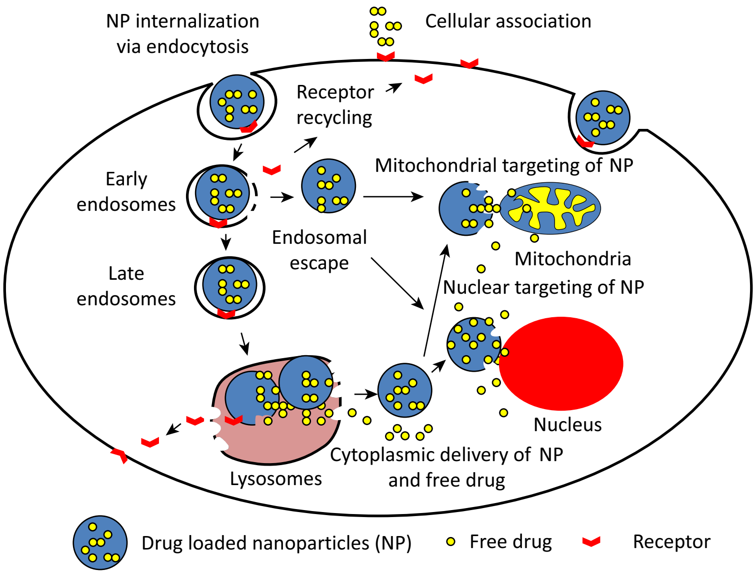

FIGURE 15.12 Endocytosis of drugs. Schematic drawing of the cytosolic delivery and organelle-specific targeting of drug-loaded nanoparticles via receptor-mediated endocytosis. After receptor-mediated cell association with nanoparticles, the nanoparticles are engulfed in a vesicle known as an early endosome. Nanoparticles formulated with an endosome disrupting property disrupt the endosomes followed by cytoplasmic delivery. On the other hand, if nanoparticles are captured in early endosomes, they may make their way to lysosomes as late endosomes where their degradation takes place. Only fraction of nondegraded drug released in the cytoplasm interacts with cellular organelles in a random fashion. However, cytosolic delivery of a fraction of organelle-targeted nanoparticles via endosomal escape or from lysosomes travels to the targeting organelles to deliver their therapeutic cargo. *Source: Çağdaş M, Sezer AD, Bucak S. Liposomes as Potential Drug Carrier Systems for Drug Delivery. Semantic Scholar. 2014. https://www.semanticscholar.org/paper/Liposomes-as-Potential-Drug-Carrier-Systems-for-%C3%87a%C4%9Fda%C5%9F-Sezer/4bf4b6f6de2c5c21dea8539c9dbb1ec0893ae5bc.*

15.2 "Target-selective" drugs

15.2.1 General

Up until the last few decades, drugs were evaluated by their ability to kill tumor cells with relative sparing of normal cells. The "targets" were thought to be general components of cells, such as DNA or proteins, and it was assumed that the same targets may be present in different kinds of cells. This was especially because of the regularity of toxic side effects, for example, on the cells of normal bone marrow, hair follicles, and nerves.

In recent decades, there has been increasing knowledge of specific molecules which are often overactive in tumor cell populations (see in Chapter 4). It has been found that different cases of the same tumor type may have different disturbances in these growth factors and progrowth pathways. Because of this, search for drugs which can selectively inhibit biochemical targets in tumor cells have been undertaken. These drugs also produce relatively fewer side effects and have been called "nontoxic" anticancer drugs [60,61].

It should be noted that the term "selective targeting" can also refer to chemical modifications of drugs so that they are only taken up into selected cells [62,63].

PARP inhibitors for patients who are BRCA-mutant are described in Section 15.1.1.

15.2.2 Antibodies against specific cell surface receptors

Antibodies were one of the earliest mechanisms of targeting drugs to specific cells [63]. Mainly, the antibodies are directed against cell type−specific surface proteins which act as receptors for signaling pathways [64−66]. They are all named with the suffix "-mab" (monoclonal antibodies).

While the principle of antibody-mediated damage to cells is well recognized, the efficacies of these agents are affected by factors related to their targeting and pharmacokinetics. These factors include that tumor-specific antigens may be difficult to identify and may only be variably expressed by the tumor cells. Furthermore, antibody conjugates, being abnormal proteins by definition, may be removed from the blood stream by the reticuloendothelial system (Appendix A1.1). In addition, the antibody recognition site may bind to the cancer cell, but endocytosis may be defective in the cancer cell, so that the conjugate may not enter the cell (see above) (Table 15.1).

Among the most extensively studied at the present time are

(i) Trastuzumab. This is an antibody against the HER-2 receptor (see Section 6.3.2) and is used for patients with breast carcinomas which express this surface protein.
(ii) Rituximab. This is an antibody against the surface receptor known as CD20. This antigen is expressed almost exclusively by B lymphocytes, and hence the drug is useful against B-cell tumorous conditions including leukemias and lymphomas, as well as in some autoimmune conditions.

15.2.3 Drugs against intracellular signaling enzymes

The importance of target-selective drugs is that there are a significant number of different targets which are suitable for blocking by drugs [60,67−70]. They mainly are named with the suffix "-ib" (Table 15.2).

To affect these targets, these drugs must enter cells, must be small, and the effect event must produce an excess of the progrowth substance, which can then be inhibited by a drug. The most established ones are imatinib for GIST tumors [71] and vemurafenib for *BRAF*-mutant late-stage melanoma [72].

Particular proteins can be inhibited, for example, the heat shock protein Hsp90 by geldanamycin, a product of *Streptomyces hygroscopicus* [73].

15.2.4 Antiangiogenesis drugs

Trials of antiangiogenic drugs have so far shown no significant beneficial effect although work is continuing in this area [74,75]. To be beneficial, the drugs would have to be given for a long as there are known to be tumor cells in the patient's body (see in Chapter 13).

15.2.5 Aptamers and aptamer targeting

Nucleic acid aptamers, often termed chemical antibodies, are short, single-stranded DNA or RNA molecules (20−100 nucleotides in length) which are three-dimensionally structured to bind to specific molecular targets in the same way that antibodies bind antigens [76].

Aptamers can serve as agonists or antagonists of specific molecules. They can also be designed to be cell type−specific, and, with an attached drug, be used in cell-targeted therapies.

15.2.6 Difficulties in drugging certain targets

Many biochemical activities depend on small "pocket" enzymatic sites. These are druggable in the sense that molecules which are large enough to block the "pocket" may also be able

TABLE 15.1 FDA-approved therapeutic monoclonal antibodies.

Antibody	Brand name	Company	Approval date	Route	Type	Target	Indication (targeted disease)	BLA STN	Drug label
Abciximab	ReoPro	Centocor	12/22/1994	Intravenous	Chimeric Fab	GPIIb/IIIa	Percutaneous coronary intervention	103575	Link
Adalimumab	Humira	Abbvie	12/31/2002	Subcutaneous	Fully human	TNF	Rheumatoid arthritis	125057	Link
Adalimumab-adbm	Cyltezo	Boehringer Ingelheim	8/25/17	Subcutaneous	Fully human, biosimilar	TNF	Rheumatoid arthritis Juvenile idiopathic arthritis Psoriatic arthritis Ankylosing spondylitis Crohn's disease Ulcerative colitis Plaque psoriasis	761058	Link
Adalimumab-atto	Amjevita	Amgen	9/23/2016	Subcutaneous	Fully human, biosimilar	TNF	Rheumatoid arthritis Juvenile idiopathic arthritis Psoriatic arthritis Ankylosing spondylitis Crohn's disease Ulcerative colitis Plaque psoriasis	761024	Link
Ado-trastuzumab emtansine	Kadcyla	Genentech	2/22/2013	Intravenous	Humanized, antibody–drug conjugate	HER2	Metastatic breast cancer	125427	Link
Alemtuzumab	Campath, Lemtrada	Genzyme	5/7/2001	Intravenous	Humanized	CD52	B-cell chronic lymphocytic leukemia	103948	Link
Alirocumab	Praluent	Sanofi Aventis	7/24/2015	Subcutaneous	Fully human	PCSK9	Heterozygous familial hypercholesterolemia Refractory hypercholesterolemia	125559	Link
Atezolizumab	Tecentriq	Genentech	5/18/2016	Intravenous	Humanized	PD-L1	Urothelial carcinoma	761034	Link
Atezolizumab	Tecentriq	Genentech	10/18/2016	Intravenous	Humanized	PD-L1	Urothelial carcinoma Metastatic nonsmall cell lung cancer	761041	Link
Avelumab	Bavencio	EMD Serono	3/23/2017	Intravenous	Fully human	PD-L1	Metastatic merkel cell carcinoma	761049	Link
Basiliximab	Simulect	Novartis	5/12/1998	Intravenous	Chimeric	IL2RA	Prophylaxis of acute organ rejection in renal transplant	103764	Link
Belimumab	Benlysta	Human genome Sciences	3/9/2011	Intravenous	Fully human	BLyS	Systemic lupus erythematosus	125370	Link
Benralizumab	Fasenra	Astrazeneca	11/14/17	Subcutaneous	Humanized	Interleukin-5 receptor alpha subunit	Severe asthma, eosinophilic phenotype	761070	Link
Bevacizumab	Avastin	Genentech	2/26/2004	Intravenous	Humanized	VEGF	Metastatic colorectal cancer	125085	Link
Bevacizumab-awwb	Mvasi	Amgen	9/14/17	Intravenous	Humanized, biosimilar	VEGF	Metastatic colorectal cancer Nonsquamous nonsmall cell lung carcinoma Glioblastoma Metastatic renal cell carcinoma Cervical cancer	761028	Link
Bezlotoxumab	Zinplava	Merck	10/21/2016	Intravenous	Fully human	Clostridium difficile toxin B	Prevent recurrence of Clostridium difficile infection	761046	Link

Blinatumomab	Blincyto	Amgen	12/3/2014	Intravenous	Mouse, bispecific	CD19	Precursor B-cell acute lymphoblastic leukemia	125557	Link
Brentuximab vedotin	Adcetris	Seattle genetics	9/19/2011	Intravenous	Chimeric, antibody—drug conjugate	CD30	Hodgkin lymphoma Anaplastic large-cell lymphoma	125388	Link
Brodalumab	Siliq	Valeant	2/15/2017	Subcutaneous	Chimeric	IL17RA	Plaque psoriasis	761032	Link
Burosumab-twza	Crysvita	Ultragenyx	4/17/18	Subcutaneous	Fully human	FGF23	X-linked hypophosphatemia	761068	Link
Canakinumab	Ilaris	Novartis	6/17/2009	Subcutaneous	Fully human	IL1B	Cryopyrin-associated periodic syndrome	125319	Link
Capromab pendetide	ProstaScint	Cytogen	10/28/1996	Intravenous	Murine, radiolabeled	PSMA	Diagnostic imaging agent in newly diagnosed prostate cancer or postprostatectomy	103608	Link
Certolizumab pegol	Cimzia	UCB (company)	4/22/2008	Subcutaneous	Humanized	TNF	Crohn's disease	125160	Link
Cetuximab	Erbitux	ImClone systems	2/12/2004	Intravenous	Chimeric	EGFR	Metastatic colorectal carcinoma	125084	Link
Daclizumab	Zenapax	Roche	12/10/1997	Intravenous	Humanized	IL2RA	Prophylaxis of acute organ rejection in renal transplant	103749	Link
Daclizumab	Zinbryta	Biogen	5/27/2016	Subcutaneous	Humanized	IL2R	Multiple sclerosis	761029	Link
Daratumumab	Darzalex	Janssen Biotech	11/16/2015	Intravenous	Fully human	CD38	Multiple myeloma	761036	Link
Denosumab	Prolia, Xgeva	Amgen	6/1/2010	Subcutaneous	Fully human	RANKL	Postmenopausal women with osteoporosis	125320	Link
Dinutuximab	Unituxin	United therapeutics	3/10/2015	Intravenous	Chimeric	GD2	Pediatric high-risk neuroblastoma	125516	Link
Dupilumab	Dupixent	Regeneron	3/28/2017	Subcutaneous	Fully human	IL4RA	Atopic dermatitis	761055	Link
Durvalumab	Imfinzi	AstraZeneca	5/1/2017	Intravenous	Fully human	PD-L1	Urothelial carcinoma	761069	Link
Eculizumab	Soliris	Alexion	3/16/2007	Intravenous	Humanized	Complement component 5	Paroxysmal nocturnal hemoglobinuria	125166	Link
Elotuzumab	Empliciti	Bristol-Myers Squibb	11/30/2015	Intravenous	Humanized	SLAMF7	Multiple myeloma	761035	Link
Emicizumab-kxwh	Hemlibra	Genentech	11/16/17	Subcutaneous	Humanized, bispecific	Factor IXa, Factor X	Hemophilia A (congenital Factor VIII deficiency) with factor VIII inhibitors.	761083	Link
Erenumab-aooe	Aimovig	Amgen	5/17/18	Subcutaneous	Fully human	CGRP receptor	Migraine headache prevention	761077	Link
Evolocumab	Repatha	Amgen	8/27/2015	Subcutaneous	Fully human	PCSK9	Heterozygous familial hypercholesterolemia Refractory hypercholesterolemia	125522	Link
Gemtuzumab ozogamicin	Mylotarg	Wyeth	9/1/17	Intravenous	Humanized, antibody—drug conjugate	CD33	Acute myeloid leukemia	761060	Link
Golimumab	Simponi	Centocor	4/24/2009	Subcutaneous	Fully human	TNF	Rheumatoid arthritis Psoriatic arthritis Ankylosing spondylitis	125289	Link
Golimumab	Simponi Aria	Janssen Biotech	7/18/2013	Intravenous	Fully human	TNF	Rheumatoid arthritis	125433	Link

Continued

TABLE 15.1 FDA-approved therapeutic monoclonal antibodies.—cont'd

Antibody	Brand name	Company	Approval date	Route	Type	Target	Indication (targeted disease)	BLA STN	Drug label
Guselkumab	Tremfya	Janssen Biotech	7/13/17	Subcutaneous	Fully human	IL23	Plaque psoriasis	761061	Link
Ibalizumab-uiyk	Trogarzo	TaiMed biologics	3/6/18	Intravenous	Humanized	CD4	HIV	761065	Link
Ibritumomab tiuxetan	Zevalin	Spectrum Pharmaceuticals	2/19/2002	Intravenous	Murine, radioimmunotherapy	CD20	Relapsed or refractory low-grade, follicular, or transformed B-cell non-Hodgkin's lymphoma	125019	Link
Idarucizumab	Praxbind	Boehringer Ingelheim	10/16/2015	Intravenous	Humanized Fab	Dabigatran	Emergency reversal of anticoagulant dabigatran	761025	Link
Infliximab	Remicade	Centocor	8/24/1998	Intravenous	Chimeric	TNF alpha	Crohn's disease	103772	Link
Infliximab-abda	Renflexis	Samsung Bioepis	4/21/2017	Intravenous	Chimeric, biosimilar	TNF	Crohn's disease Ulcerative colitis Rheumatoid arthritis Ankylosing spondylitis Psoriatic arthritis Plaque psoriasis	761054	Link
Infliximab-dyyb	Inflectra	Celltrion Healthcare	4/5/2016	Intravenous	Chimeric, biosimilar	TNF	Crohn's disease Ulcerative colitis Rheumatoid arthritis Ankylosing spondylitis Psoriatic arthritis Plaque psoriasis	125544	Link
Infliximab-qbtx	Ixifi	Pfizer	12/13/17	Intravenous	Chimeric, biosimilar	TNF	Crohn's disease Ulcerative colitis Rheumatoid arthritis Ankylosing spondylitis Psoriatic arthritis Plaque psoriasis	761072	Link
Inotuzumab ozogamicin	Besponsa	Wyeth	8/17/17	Intravenous	Humanized, antibody–drug conjugate	CD22	Precursor B-cell acute lymphoblastic leukemia	761040	Link
Ipilimumab	Yervoy	Bristol-Myers Squibb	3/25/2011	Intravenous	Fully human	CTLA-4	Metastatic melanoma	125377	Link
Ixekizumab	Taltz	Eli Lilly	3/22/2016	Subcutaneous	Humanized	IL17A	Plaque psoriasis	125521	Link
Mepolizumab	Nucala	GlaxoSmithKline	11/4/2015	Subcutaneous	Humanized	IL5	Severe asthma	125526	Link
Natalizumab	Tysabri	Biogen Idec	11/23/2004	Intravenous	Humanized	Alpha-4 integrin	Multiple sclerosis	125104	Link
Necitumumab	Portrazza	Eli Lilly	11/24/2015	Intravenous	Fully human	EGFR	Metastatic squamous nonsmall cell lung carcinoma	125547	Link
Nivolumab	Opdivo	Bristol-Myers Squibb	12/22/2014	Intravenous	Fully human	PD-1	Metastatic melanoma	125554	Link
Nivolumab	Opdivo	Bristol-Myers Squibb	3/4/2015	Intravenous	Fully human	PD-1	Metastatic squamous nonsmall cell lung carcinoma	125527	Link
Obiltoxaximab	Anthem	Elusys therapeutics	3/18/2016	Intravenous	Chimeric	Protective antigen of the anthrax toxin	Inhalational anthrax	125509	Link

Obinutuzumab	Gazyva	Genentech	11/1/2013	Intravenous	Humanized	CD20	Chronic lymphocytic leukemia	125486	Link
Ocrelizumab	Ocrevus	Genentech	3/28/2017	Intravenous	Humanized	CD20	Multiple sclerosis	761053	Link
Ofatumumab	Arzerra	Glaxo Grp	10/26/2009	Intravenous	Fully human	CD20	Chronic lymphocytic leukemia	125326	Link
Olaratumab	Lartruvo	Eli Lilly	10/19/2016	Intravenous	Fully human	PDGFRA	Soft tissue sarcoma	761038	Link
Omalizumab	Xolair	Genentech	6/20/2003	Subcutaneous	Humanized	IgE	Moderate to severe persistent asthma	103976	Link
Palivizumab	Synagis	MedImmune	6/19/1998	Intramuscular	Humanized	F Protein of RSV	Respiratory syncytial virus	103770	Link
Panitumumab	Vectibix	Amgen	9/27/2006	Intravenous	Fully human	EGFR	Metastatic colorectal cancer	125147	Link
Pembrolizumab	Keytruda	Merck	9/4/2014	Intravenous	Humanized	PD-1	Metastatic melanoma	125514	Link
Pertuzumab	Perjeta	Genentech	6/8/2012	Intravenous	Humanized	HER2	Metastatic breast cancer	125409	Link
Ramucirumab	Cyramza	Eli Lilly	4/21/2014	Intravenous	Fully human	VEGFR2	Gastric cancer	125477	Link
Ranibizumab	Lucentis	Genentech	6/30/2006	Intravitreal injection	Humanized	VEGFR1 VEGFR2	Wet age−related macular degeneration	125156	Link
Raxibacumab	Raxibacumab	Human genome Sciences	12/24/2012	Intravenous	Fully human	Protective antigen of Bacillus anthracis	Inhalational anthrax	125349	Link
Reslizumab	Cinqair	Teva	3/23/2016	Intravenous	Humanized	IL5	Severe asthma	761033	Link
Rituximab	Rituxan	Genentech	11/26/1997	Intravenous	Chimeric	CD20	B-cell non-Hodgkin's lymphoma	103705	Link
rituximab and hyaluronidase	Rituxan Hycela	Genentech	6/22/17	Subcutaneous	Chimeric, coformulated	CD20	Follicular lymphoma Diffuse large B-cell lymphoma Chronic lymphocytic leukemia	761064	Link
Sarilumab	Kevzara	Sanofi Aventis	5/22/17	Subcutaneous	Fully human	IL6R	Rheumatoid arthritis	761037	Link
Secukinumab	Cosentyx	Novartis	1/21/2015	Subcutaneous	Fully human	IL17A	Plaque psoriasis	125504	Link
Siltuximab	Sylvant	Janssen Biotech	4/23/2014	Intravenous	Chimeric	IL6	Multicentric Castleman's disease	125496	Link
Tildrakizumab-asmn	Ilumya	Merck	3/20/18	Subcutaneous	Humanized	IL23	Plaque psoriasis	761067	Link
Tocilizumab	Actemra	Genentech	1/8/2010	Intravenous	Humanized	IL6R	Rheumatoid arthritis	125276	Link
Tocilizumab	Actemra	Genentech	10/21/2013	Intravenous Subcutaneous	Humanized	IL6R	Rheumatoid arthritis Polyarticular juvenile idiopathic arthritis Systemic juvenile idiopathic arthritis	125472	Link
Trastuzumab	Herceptin	Genentech	9/25/1998	Intravenous	Humanized	HER2	Metastatic breast cancer	103792	Link
Trastuzumab-dkst	Ogivri	Mylan	12/1/17	Intravenous	Humanized, biosimilar	HER2	HER2-overexpressing breast cancer, metastatic gastric or gastroesophageal junction adenocarcinoma	761074	Link
Ustekinumab	Stelara	Centocor	9/25/2009	Subcutaneous	Fully human	IL12 IL23	Plaque psoriasis	125261	Link
Ustekinumab	Stelara	Janssen Biotech	9/23/2016	Subcutaneous Intravenous	Fully human	IL12 IL23	Plaque psoriasis Psoriatic arthritis Crohn's disease	761044	Link
Vedolizumab	Entyvio	Takeda	5/20/2014	Intravenous	Humanized	integrin receptor	Ulcerative colitis Crohn's disease	125476	Link

TABLE 15.2 Some commonly used selective anticancer drugs. *From*

The FDA has approved targeted therapies for the treatment of some patients with the following types of cancer (some targeted therapies have been approved to treat more than one type of cancer):

Adenocarcinoma of the stomach or gastroesophageal junction: Trastuzumab (Herceptin), ramucirumab (Cyramza)

Bladder cancer: Atezolizumab (Tecentriq), nivolumab (Opdivo), durvalumab (Imfinzi), avelumab (Bavencio), pembrolizumab (Keytruda)

Brain cancer: Bevacizumab (Avastin), everolimus (Afinitor)

Breast cancer: Everolimus (Afinitor), tamoxifen (Nolvadex), toremifene (Fareston), Trastuzumab (Herceptin), fulvestrant (Faslodex), anastrozole (Arimidex), exemestane (Aromasin), lapatinib (Tykerb), letrozole (Femara), pertuzumab (Perjeta), ado-trastuzumab emtansine (Kadcyla), palbociclib (Ibrance), ribociclib (Kisqali), neratinib maleate (Nerlynx), abemaciclib (Verzenio), olaparib (Lynparza), atezolizumab (Tecentriq)

Cervical cancer: Bevacizumab (Avastin), pembrolizumab (Keytruda)

Colorectal cancer: Cetuximab (Erbitux), panitumumab (Vectibix), bevacizumab (Avastin), ziv-aflibercept (Zaltrap), regorafenib (Stivarga), ramucirumab (Cyramza), nivolumab (Opdivo), ipilimumab (Yervoy)

Dermatofibrosarcoma protuberans: Imatinib mesylate (Gleevec)

Endocrine/neuroendocrine tumors: Lanreotide acetate (Somatuline Depot), avelumab (Bavencio), lutetium Lu 177-dotatate (Lutathera), iobenguane I 131 (Azedra)

Head and neck cancer: Cetuximab (Erbitux), pembrolizumab (Keytruda), nivolumab (Opdivo)

Gastrointestinal stromal tumor: Imatinib mesylate (Gleevec), sunitinib (Sutent), regorafenib (Stivarga)

Giant cell tumor of the bone: Denosumab (Xgeva)

Kidney cancer: Bevacizumab (Avastin), sorafenib (Nexavar), sunitinib (Sutent), pazopanib (Votrient), temsirolimus (Torisel), everolimus (Afinitor), axitinib (Inlyta), nivolumab (Opdivo), cabozantinib (Cabometyx), lenvatinib mesylate (Lenvima), ipilimumab (Yervoy), pembrolizumab (Keytruda)

Leukemia: Tretinoin (Vesanoid), imatinib mesylate (Gleevec), dasatinib (Sprycel), nilotinib (Tasigna), bosutinib (Bosulif), rituximab (Rituxan), alemtuzumab (Campath), ofatumumab (Arzerra), obinutuzumab (Gazyva), ibrutinib (Imbruvica), idelalisib (Zydelig), blinatumomab (Blincyto), venetoclax (Venclexta), ponatinib hydrochloride (Iclusig), midostaurin (Rydapt), enasidenib mesylate (Idhifa), inotuzumab ozogamicin (Besponsa), tisagenlecleucel (Kymriah), gemtuzumab ozogamicin (Mylotarg), rituximab and hyaluronidase human (Rituxan Hycela), ivosidenib (Tibsovo), duvelisib (Copiktra), moxetumomab pasudotox-tdfk (Lumoxiti), glasdegib maleate (Daurismo), gilteritinib (Xospata), tagraxofusp-erzs (Elzonris)

Liver cancer: Sorafenib (Nexavar), regorafenib (Stivarga), nivolumab (Opdivo), lenvatinib mesylate (Lenvima), pembrolizumab (Keytruda), cabozantinib (Cabometyx)

Lung cancer: Bevacizumab (Avastin), crizotinib (Xalkori), erlotinib (Tarceva), gefitinib (Iressa), afatinib dimaleate (Gilotrif), ceritinib (LDK378/Zykadia), ramucirumab (Cyramza), nivolumab (Opdivo), pembrolizumab (Keytruda), osimertinib (Tagrisso), necitumumab (Portrazza), alectinib (Alecensa), atezolizumab (Tecentriq), brigatinib (Alunbrig), trametinib (Mekinist), dabrafenib (Tafinlar), durvalumab (Imfinzi), dacomitinib (Vizimpro), lorlatinib (Lorbrena)

Lymphoma: Ibritumomab tiuxetan (Zevalin), denileukin diftitox (Ontak), brentuximab vedotin (Adcetris), rituximab (Rituxan), vorinostat (Zolinza), romidepsin (Istodax), bexarotene (Targretin), bortezomib (Velcade), pralatrexate (Folotyn), ibrutinib (Imbruvica), siltuximab (Sylvant), idelalisib (Zydelig), belinostat (Beleodaq), obinutuzumab (Gazyva), nivolumab (Opdivo), pembrolizumab (Keytruda), rituximab and hyaluronidase human (Rituxan Hycela), copanlisib hydrochloride (Aliqopa), axicabtagene ciloleucel (Yescarta), acalabrutinib (Calquence), tisagenlecleucel (Kymriah), venetoclax (Venclexta), mogamulizumab-kpkc (Poteligeo), duvelisib (Copiktra)

TABLE 15.2 Some commonly used selective anticancer drugs. *From*—cont'd

Microsatellite instability-high or mismatch repair-deficient solid tumors: Pembrolizumab (Keytruda)
Multiple myeloma: Bortezomib (Velcade), carfilzomib (Kyprolis), panobinostat (Farydak), daratumumab (Darzalex), ixazomib citrate (Ninlaro), elotuzumab (Empliciti)
Myelodysplastic/myeloproliferative disorders: Imatinib mesylate (Gleevec), ruxolitinib phosphate (Jakafi)
Neuroblastoma: Dinutuximab (Unituxin)
Ovarian epithelial/fallopian tube/primary peritoneal cancers: Bevacizumab (Avastin), olaparib (Lynparza), rucaparib camsylate (Rubraca), niraparib tosylate monohydrate (Zejula)
Pancreatic cancer: Erlotinib (Tarceva), everolimus (Afinitor), sunitinib (Sutent)
Prostate cancer: Cabazitaxel (Jevtana), enzalutamide (Xtandi), abiraterone acetate (Zytiga), radium 223 dichloride (Xofigo), apalutamide (Erleada)
Skin cancer: Vismodegib (Erivedge), sonidegib (Odomzo), ipilimumab (Yervoy), vemurafenib (Zelboraf), trametinib (Mekinist), dabrafenib (Tafinlar), pembrolizumab (Keytruda), nivolumab (Opdivo), cobimetinib (Cotellic), alitretinoin (Panretin), avelumab (Bavencio), encorafenib (Braftovi), binimetinib (Mektovi), cemiplimab-rwlc (Libtayo)
Soft tissue sarcoma: Pazopanib (Votrient), alitretinoin (Panretin)
Solid tumors with an *NTRK* gene fusion: Larotrectinib sulfate (Vitrakvi)
Stomach cancer: Pembrolizumab (Keytruda)
Systemic mastocytosis: Imatinib mesylate (Gleevec), midostaurin (Rydapt)
Thyroid cancer: Cabozantinib (Cometriq), vandetanib (Caprelsa), sorafenib (Nexavar), lenvatinib mesylate (Lenvima), trametinib (Mekinist), dabrafenib (Tafinlar)

Source: https://www.cancer.gov/about-cancer/treatment/types/targeted-therapies/targeted-therapies-fact-sheet (April 2019).

to enter cells. However, other biochemical activities depend on large areas of interaction of proteins, especially involving weak intermolecular bonds. These sites may be too large to be affected by small molecules attaching to only a small part of their overall dimensions. For some biochemical actions, this problem may be overcome by a small molecule which binds a secondary site on the target so that a conformational change is caused [77] (Fig. 15.13).

15.2.7 "On-target" and "off-target" effects of targeted drugs

"On-target side effect" refers to undesired excessive effects of drugs on the target cells. "Off-target side effect" applies to any adverse effects on nontarget cells, in relation to any of their biochemical structures or processes [78].

These can arise in many ways, for example, enhancing or reducing the actions of another drug, unexpected rapid or slow accumulation and toxicity in cells, unexpected metabolic products affecting cells, and allergic reactions [79].

15.3 Aspects of personalized medicine

15.3.1 Terminology

In oncology, "personalized medicine" is used in three contexts [80–83]:

(i) Examining the tumor for characteristics which may assist in choices of the most particularly appropriate treatment for the particular patient. The histologic grading and molecular tests of tumor samples (Chapter 10) are in this category.

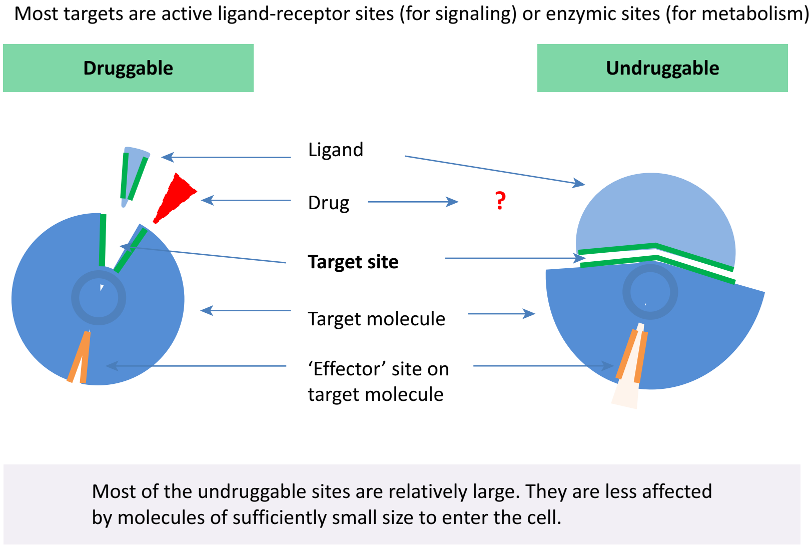

FIGURE 15.13 Druggable and undruggable targets.

(ii) Assessing the patient's normal biochemical and genomic characteristics to establish which forms of therapy may be most effective in them, and/or for which they may suffer the least side effects. This can include comorbidities and concurrent therapies.

(iii) Determining in currently healthy individuals, the risks of future disease and hence the steps which might be taken to avoid these illnesses, i.e., "personalized disease prevention," as described in Section 8.7.7.

15.3.2 Studies of the patient's tumor cells in cultures or as xenografts

Assessing patients' tumors for genomic abnormalities suitable for specific treatment are discussed in Section 10.2. Two further types of assessments are mentioned here.

(a) Cell biological and pharmacological assessments of cultured cells from the patient's tumor

Generally, attempts to grow patients' tumor cells in culture and use them to assess their drug sensitivities—analogous to those for testing bacterial sensitivities to antibiotics—have not been successful. The main difficulty has been that few cell lines suitable for the purpose can be grown from most cases of tumors. This is especially because of only small amounts of tissue which might be available from certain sorts of biopsy such as needle biopsies [84].

In studies in which cultures of cell lines have been successful, there has been little correlation between the sensitivity to anticancer agents of cells grown from a tumor and the "responsiveness" (shrinkage) of the tumor in the patient when the same agent(s) is administered.

Some experience has been gained with attempts to xenograft tumor pieces into immunodeficient animals, but this is still evolving [85].

Reasons may include the following:

(i) The microenvironmental situation of tumor cells—and hence pretarget resistance factors/pharmacodynamics (see Section 13.1.2 and Appendix 8)—in the whole patient or a site in the body is chemically and biologically more complex than in the media used for in vitro culture or the laboratory animal.

(ii) In tumor lines which are established, the proliferating cells in the tumor are heterogeneously heterogenizing for cell kinetics and for sensitivity to anticancer agents, as well as for many other properties (see Sections 2.8 and 13.2). The assumption that particular cell cultures from a tumor are representative of all the proliferating subpopulations (present and future) in the tumor mass as a whole is fundamentally unwarranted [86,87]. Hence, even if the factors above are not acting in a particular tumor, the proliferating cells in the tumor masses in a patient may well produce new mutants of with different properties by the time the studies of the sensitivities of the cell line cultured earlier become available. This is particularly true of the development of more hyperploid cell types being associated with increasing numbers of copies of defense genes against anticancer drugs.

15.3.3 Patients' normal genomes and therapy (pharmacogenomics)

Another aspect of personalized medicine is attempting to assess the likelihood of a particular side effect of an agent in the individual person on the basis of their genome [88–90]. As is shown in Appendix 9, potential drugs are tested by "Phase 1" trials for unacceptable side effects. Hence, genomic variations which do not fail the Phase 1 trials by causing these side effects should not be very common in the general population.

Most prescription drugs are metabolized in the body by the cytochrome p450 family of enzymes (see Section 12.2.2). Different enzymes metabolize different drugs. An individual might have excess metabolizing capacity, in which case the drug will be ineffective. On the other hand, another individual might be deficient metabolizing the drug, in which case, effects of overdosing may occur.

With the ability to sequence large amounts of DNA (see Appendix 2), it has also been suggested that the information about the patient's genome may assist in avoiding the idiosyncratic (rare, individual) side effects of drugs. This has been achieved for some drugs, but few reactions to anticancer drugs have been clarified in this way [91].

15.4 Chemotherapies for particular malignant tumors

The particular regimen of chemotherapy—with or without other concurrent therapies—received by a patient for a particular tumor depends on the stage of the cancer, and the patient's particular features, especially, age, comorbidities, allergies, and preferences concerning possible side effects. This can include "off-label" use of cytotoxic drugs, which is the use of a marketed drug outside the conditions described in the summary of product conditions. In oncology, this may occur in patients with other tumors expressing the same target [92].

The following are summaries of guidelines from the American Society of Clinical Oncology–sponsored website Cancer.Net.

15.4.1 Small-celled carcinoma of lung

Primary treatment is usually chemotherapy consisting of a topoisomerase inhibitor (etoposide or irinotecan) plus a platinum-based drug such as cis-platin or carboplatin.

For patients with limited stage, small cell lung cancer, radiation therapy is combined with chemotherapy (see above). Radiation therapy is best when given during the first or second month of chemotherapy [93].

15.4.2 Nonsmall-celled carcinoma of lung

This group can be divided into adenocarcinomas, squamous cell carcinomas, and undifferentiated carcinomas. The World Health Organization Union for International Cancer control produced a Review of Cancer Medicines for use with nonsmall cell lung cancer in 2014 [94] and in 2015 published the classification of lung tumors which was comprehensively analyzed by Travis et al. in that year [95].

Nonsmall cell carcinomas can be treated with surgery (see in Chapter 12), radiation therapy (Chapter 14), primarily cytotoxic chemotherapy, targeted chemotherapy, and immunotherapy (Chapter 16), or combinations of these (Chapter 13). Anticancer drugs are usually given in combinations. Commonly used drugs include taxanes, gemcitabine (a nucleotide analogue), pemetrexed (an antifolate), and a vinca alkaloid derivative vinorelbine, often in conjunction with a platinum-based drug.

Targeted therapies are mainly directed against angiogenesis (see in Chapter 2), the epidermal growth factor, and certain genes when mutants, e.g., anaplastic lymphoma kinase and BRAF [96].

15.4.3 Colorectal carcinoma

Chemotherapy may be recommended before or after surgery, sometimes with radiation therapy, depending on the local spread and the patient's condition (age and comorbidities).

Some common treatment regimens using these drugs include the following:

> 5-FU (TS inhibitor, hence DNA synthesis inhibitor) often in combination with folinic acid, and a telomerase inhibitor, platinum-based drug, or a targeted therapy: against angiogenesis, epidermal growth factor, or a check-point inhibitor (see in Chapter 16) [97].

15.4.4 Carcinoma of the breast

This disease includes many different in situ and invasive subtypes (see in Chapter 10) and can be treated by surgery, radiotherapy, and hormonal agents as well as chemotherapy. A wide range of drugs may be used including TS inhibitors, nucleoside analogues, taxanes, DNA cross-linking, antibiotic-derived anthracyclines, and platinum-based drugs. These are commonly in combination with other modalities. The list of current drugs approved for breast cancer can be found on the National Cancer Institute website [98].

Targeted therapies mainly involve mutant HER-2 gene and hormonal therapies (see below) [99].

15.4.5 Prostate

This disease is mainly treated by hormonal therapies (see below). For metastatic disease, a combination of a taxane and corticosteroid may be given [100].

15.4.6 Other

(a) Melanoma

(i) Stage II and III: The US Food and Drug Administration (FDA) has approved four adjuvant immunotherapies for Stage II and Stage III melanoma: high-dose interferon alfa-2b (Intron A), pegylated interferon alfa-2b (Sylatron), ipilimumab (Yervoy), and nivolumab (Opdivo).

(ii) Unresectable Stage III and Stage IV and immunotherapies and BRAF, MEK and KIT inhibitors, conventional cytotoxic therapies are second- or third-line therapies.

Dacarbazine (DTIC-Dome) is the only FDA-approved chemotherapy for melanoma. Malignant melanomas are not sensitive to others but can be treated with adjuvant immunotherapies, e.g., IFNα2b(INTRON A) which is also approved by the US FDA [101].

Other cytotoxic regimens are "off-label."

For details of drug therapies of particular tumor types, see in Refs. [102–105].

(b) Renal cell carcinoma

Current first-line therapies are targeted or immunotherapies. No conventional cytotoxic drugs are widely used.

(c) Gastric and pancreatic cancer

The range of cytotoxic therapies are similar to those for colorectal cancer.

15.5 Antihormone therapies

These are essentially limited to tumors of the secondary sex organs (breast and endometrium; prostate). The aim is to reduce the proliferation of tumor cells. The cells are not killed. The therapies consist of reductions of the production of natural hormones or the administration of antihormones.

15.5.1 Breast carcinoma

The earliest hormonal treatment for carcinoma of the breast was to reduce estrogen levels in the blood by removing the patients' ovaries. The treatment was helpful in a proportion of cases, but present therapies for reduction in estrogen stimulation to tumor cells usually involve antiestrogen drugs, such as tamoxifen. These drugs can also be used in the prevention of carcinomas of the breast and endometrium in high-risk individuals (see in Chapter 12).

(a) Estrogen production suppressors

(i) Chemical ovariectomy

In early years, cytotoxic drugs were given for this purpose, with corresponding menopause-associated side effects (see Section 13.1.5). More recently, luteinizing hormone–releasing hormone (LHRH) analogues have been used. These drugs are used more often than surgical ovariectomy. They inhibit the hormone which stimulates estrogen production in the ovaries. Symptoms of menopause may occur. Common LHRH drugs include goserelin (Zoladex) and leuprolide (Lupron), alone or with other hormone drugs (tamoxifen, aromatase inhibitors, fulvestrant) as hormone therapy in premenopausal women.

(ii) Aromatase inhibitors

Aromatase (estrogen synthetase) is found in the granulosa cells of the ovaries, and various nongonadal tissues. Aromatase inhibitors (AIs—letrozole, anastrozole, exemestane) are most useful in postmenopausal patients, although they can also be used in premenopausal women in combination with, or alternating with, ovarian suppression (e.g., tamoxifen, see below). They are taken for as long as recurrent breast carcinoma seems likely (e.g., 10 years).

(b) Estrogen receptor blockers

The main drug is tamoxifen, which blocks estrogen receptors on breast cancer cells. It inhibits growth of breast cells but increases growth of other tissues such as endometrium, even in menopausal women, being associated with endometrial carcinoma. Because of this, it is called a selective estrogen receptor modulator [106].

Tamoxifen can be used to treat existing invasive breast cancer, to prevent existing noninvasive breast cancer progressing to invasion, and in patients with a strong family history or BRCA gene positivity, as a preventative measure against developing breast tumor.

The side effects are similar to those of menopause: hot flashes, vaginal dryness or discharge, and mood swings.

(c) Antiestrogen: estrogen receptor degrader

Only a few drugs in this category are available (fulvestrant). They act by binding the estrogen receptor so that the complex is inactive and destroyed. The possibility of incorporating the drug into nanoparticles with anti-FERs antibodies has been described [107] (Fig. 15.14).

15.5.2 Prostate carcinoma

For carcinoma of the prostate, therapies include removing natural sources of androgens by bilateral orchidectomy or antiandrogenic drugs. In the past, estrogens, such as stilboestrol,

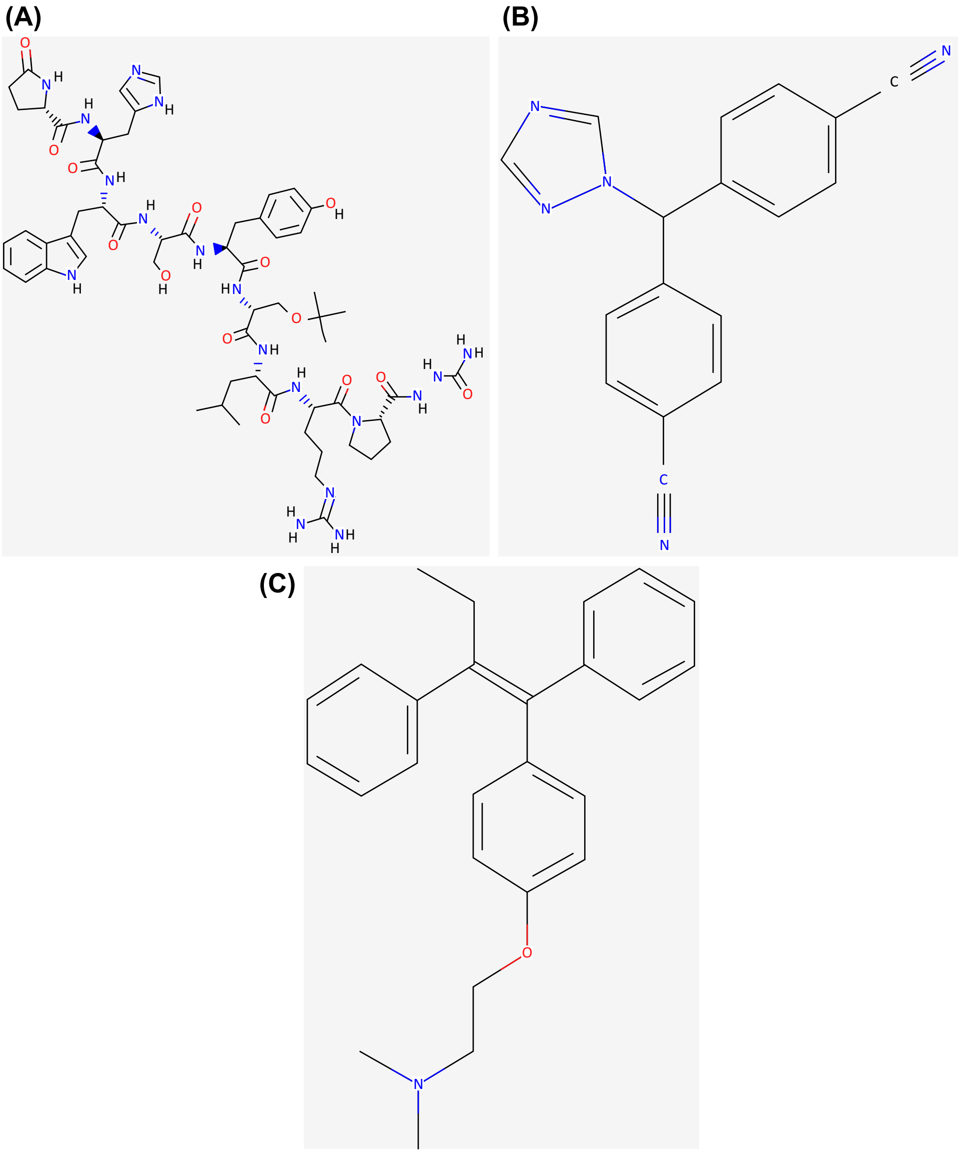

FIGURE 15.14 Structures of some female sex hormonal drugs. (A) Goserelin. PubChem Compound Database. (B) Letrozole. (C) Tamoxifen. *Source: (A)* *https://pubchem.ncbi.nlm.nih.gov/.*

were used, but the side effects of including vascular thrombosis and breast enlargement have made these drugs less popular.

Presently, antiandrogens drugs are the preferred therapies for this hormonal effect [108].

(a) Androgen production suppressants

(i) LHRH agonists (also called LHRH analogues or GnRH agonists). They lower the amount of testosterone made by the testicles. Treatment with these drugs is

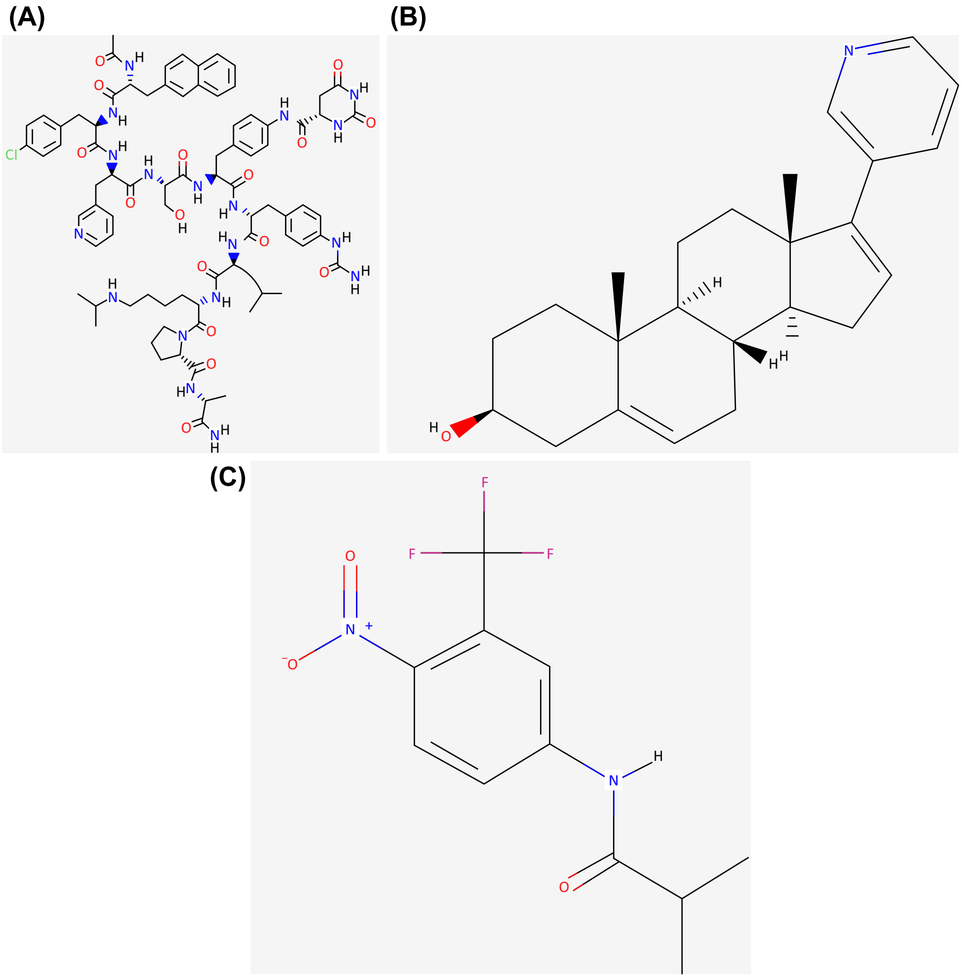

FIGURE 15.15 Structures of some male-sex hormonal drugs. (A) Degarelix. (B) Abiretarone. (C) Flutamide. *Source: PubChem Compound Database; https://pubchem.ncbi.nlm.nih.gov/ [Accessed April 2019].*

sometimes called chemical castration or medical castration because they lower androgen levels just as well as orchiectomy [109] (Fig. 15.15).

(ii) LHRH antagonists (e.g., Degarelix (Firmagon)). They work like the LHRH agonists but lower testosterone levels more quickly and does not cause tumor flare like the LHRH agonists do. Treatment with this drug can also be considered a form of medical castration.

(iii) CYP17 inhibitors (e.g., Abiraterone). These drugs target nontesticular cells, including prostate cancer cells from making their normal, low amounts of androgens. The drug suppresses corticosteroid synthesis, so that prednisone is required as a substitute.

(b) Androgen receptor blockers/antagonists (antiandrogens)

These drugs (e.g., flutamide) act on androgen receptors in the prostate. Antagonism of androgen receptors in prostate tissue reduces growth stimulation of prostatic cancer because of their action on modifying transcription factors related to cellular growth [110]. They are usually a second-line or third-line agent if orchiectomy or an LHRH agonist or antagonist no longer works by themselves. An antiandrogen is also sometimes given for a few weeks when an LHRH agonist is first started to prevent a tumor flare.

Side effects:

Reduced or absent sexual desire
Erectile dysfunction (impotence)
Shrinkage of testicles and penis
Hot flashes, which may get better or go away with time
Breast tenderness and growth of breast tissue
Osteoporosis (bone thinning), which can lead to broken bones
Anemia (low red blood cell counts)
Decreased mental sharpness
Loss of muscle mass
Weight gain
Fatigue
Increased cholesterol levels
Depression

15.6 Summary of translational issues

15.6.1 Pharmacokinetics and -dynamics in relation to tumor cells

- In vivo efficacy and safety of cancer drugs [111,112].
- Exposure—toxicity relationship to drugs (side effects including DNA damage) [113].
- Biologically dubious "lumping" of tumor types [114].

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